CSC 648 (Section 3)
Vibe Check: Milestone 1
ARKHA (Team 6)
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https://github.com/sfsu-joseo/csc648-848-project-sp26-ARKHA

Revision	Description
v1	Created document 2/7 @5:22pm, added tech stack for Care Companion and Vibe Check apps
v2	2/24 @ 9:03 pm: Finished Milestone 1, checkpoint 3 requirements

Table of Contents

Executive Summary

What's the Product?

Vibe Check is a real-time, location-based bulletin board application that helps users understand the current atmosphere, or "vibe," of nearby places. Through short, crowd-sourced notes, users can quickly see whether a location is crowded, quiet, active, or slow. Instead of relying on outdated reviews or historical averages, Vibe Check focuses on live, user-generated updates that reflect what is happening in the moment.

What Problem Are We Solving?

Existing platforms such as Google Maps and Yelp primarily rely on historical trends and past reviews to estimate how busy or active a location might be. While useful, this information does not accurately reflect real-time conditions. As a result, people often waste time and effort traveling to places that are overcrowded, noisy, closed, or otherwise unsuitable for their immediate needs. Vibe Check addresses this gap by providing timely, location-specific updates that help users make better decisions before they arrive.

Who Is the Product For?

Vibe Check is designed for individuals who frequently make spontaneous or time-sensitive decisions about where to go. This includes university students searching for study spots, busy professionals looking for productive environments, tourists exploring unfamiliar areas, event attendees checking venue conditions, and nature enthusiasts assessing trail or park activity. The platform supports anyone who benefits from real-time situational awareness.

Value Proposition

Vibe Check enables users to avoid long waits, reduce exposure to overcrowded or noisy environments, and quickly choose the right place for their needs. By presenting clear, live updates in a simple format, the application helps users save time, reduce frustration, and make confident decisions without guesswork.

What Makes Our Product Unique?

Vibe Check differentiates itself through its location-locked posting system, which ensures that notes can only be submitted and viewed within a defined geographic radius. This increases reliability and prevents irrelevant or misleading content. Additionally, preset expiration times automatically remove outdated notes, maintaining data freshness. A live heatmap further enhances visibility by showing overall activity levels in an area at a glance.

Future Potential

Looking ahead, Vibe Check has opportunities for business partnerships with cafés, venues, and local establishments that may benefit from real-time engagement insights. Gamification features such as badges and contribution streaks could encourage consistent user participation. The platform may also introduce personalized notifications and trend summaries to enhance user experience and long-term engagement.

Main Use Cases

Checking a Busy Restaurant Before Committing

Actors: John (Registered User), Vibe Check (System)

Assumptions:

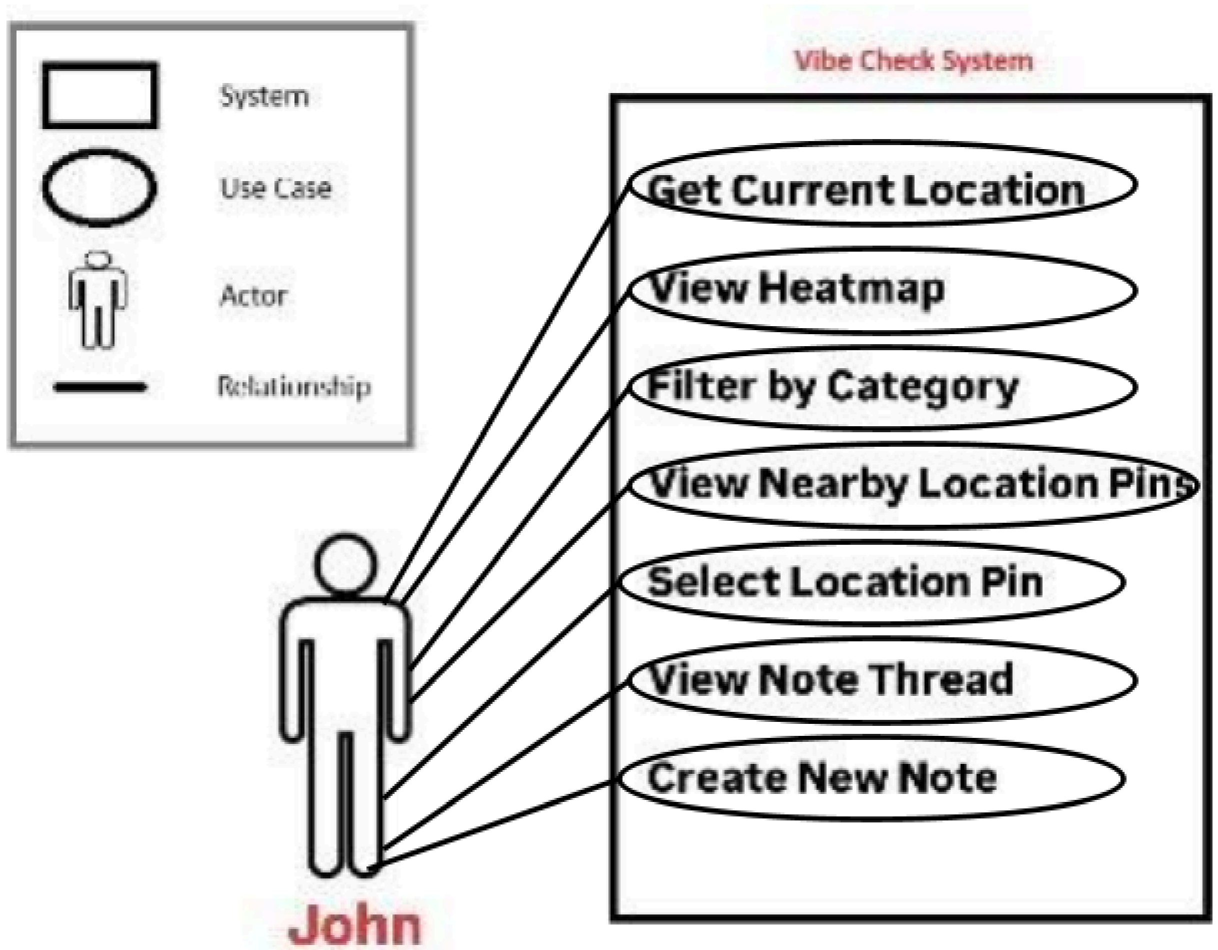
- John has access to an Internet connected device
- John has location services enabled on his device
- There are existing notes in the area created recently

Use Case: 1

John is walking around his neighborhood and is looking for a place to eat. He has to be back at work soon and does not want to waste time walking to a restaurant only to find out that there is a long wait. He opens Vibe Check and sees a heatmap of the neighborhood that he is in. He filters the heatmap by ‘Restaurants’ and selects a restaurant that is of interest to him. He taps on the restaurant pin and reads a recent note that says, “The line is going around the block! It’s about a 30 minute wait just to get inside.” John reads the replies to this note and sees other users confirming the update. John then taps on another restaurant that has a note saying, “Quick service, got great food in less than 15 minutes.” John also notices a few people reacted positively to this note. John feels more confident in making the decision to go to the second restaurant and in being able to get back to work on time because the information is current and from people who were actually at that location recently. After eating, John creates his own note on the restaurant in Vibe Check that reflects his experience, allowing future users to also get up to date information on the location.

Benefits for John:

- John can make an informed decision on how and where to spend his time without any guesswork.
- John can get a crowdsourced opinion on a location he has not been to before.



Finding a Quiet Study Spot on Campus

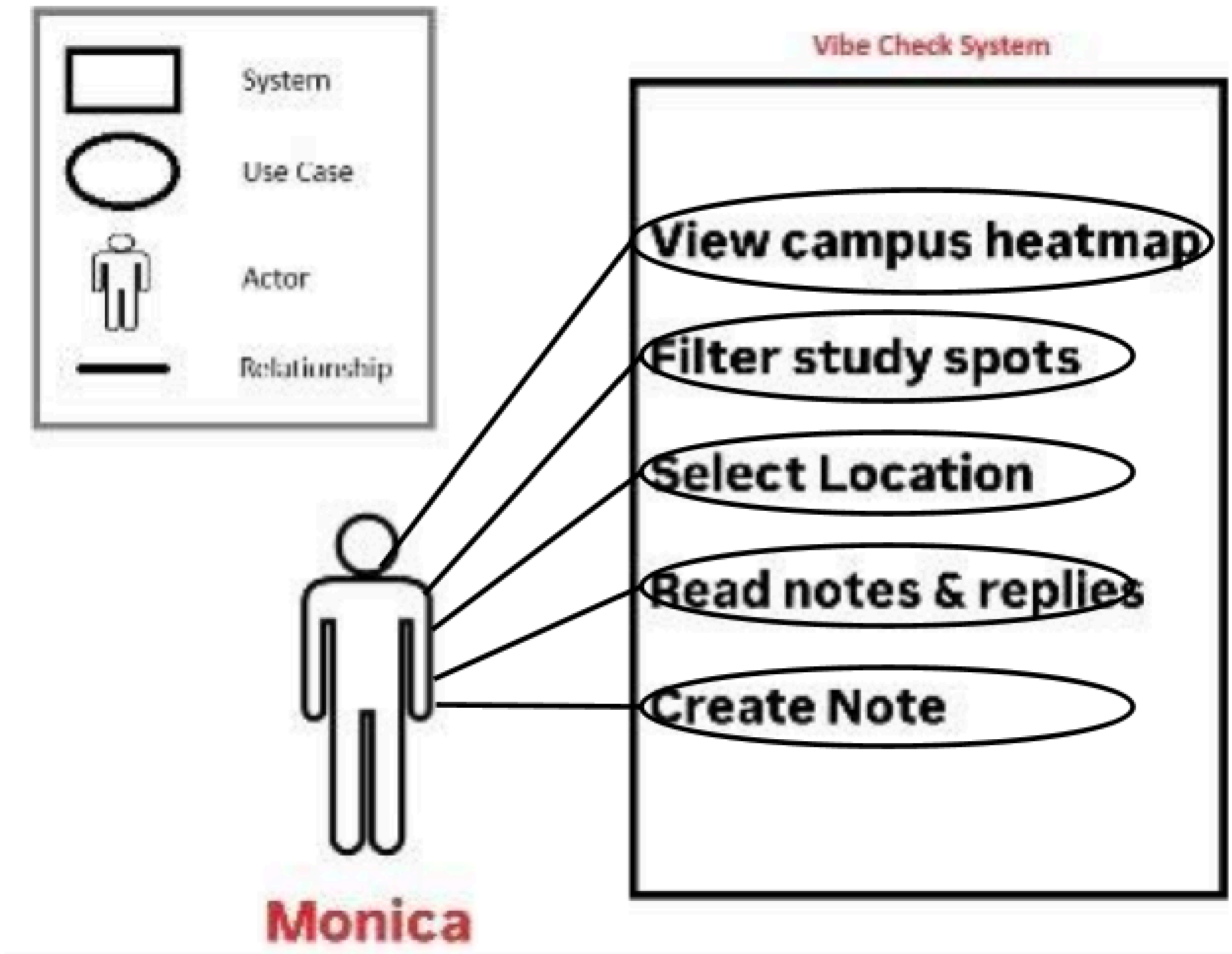
Actors:
Monica (Registered User, University Student), Vibe Check (System)

Assumptions:
- Monica has access to an Internet connected device
- Monica has location services enabled on her device
- There are existing notes in the campus area created recently

Use Case: 2
Monica is a university student preparing for her midterm exams & needs a quiet place to study for a few hours. She doesn't want to walk to the library only to find it's packed and noisy. Monica opens Vibe check & views the heatmap of her campus area. She filters the heatmap by 'Study spots' and notices several locations with varying levels of activity. She taps on the main library and reads a note posted 10 minutes ago that says, "Every seat is taken & it's pretty loud here with group projects going on". Monica checks the replies & sees multiple users confirming the crowded conditions. She then selects a smaller study lounge in another building & reads a recent note: "Super

quiet right now, plenty of empty tables by the windows." The note has several positive reactions from other students. Confident in her choice, Monica heads to the study lounge & successfully finds a peaceful spot to study. After settling in, she creates her own note on Vibe Check confirming the quiet atmosphere and available seating, helping other students who might be searching for study spots later.

Benefits for Monica:
- Monica saves time by avoiding crowded or unsuitable study locations
- Monica can find the ideal environment for her current needs (quiet, available seating)
- Monica contributes real-time information that helps fellow students make better decisions



Reviewing Places by Popularity

Actors: Lex (Registered User, Worker), Vibe Check (System)

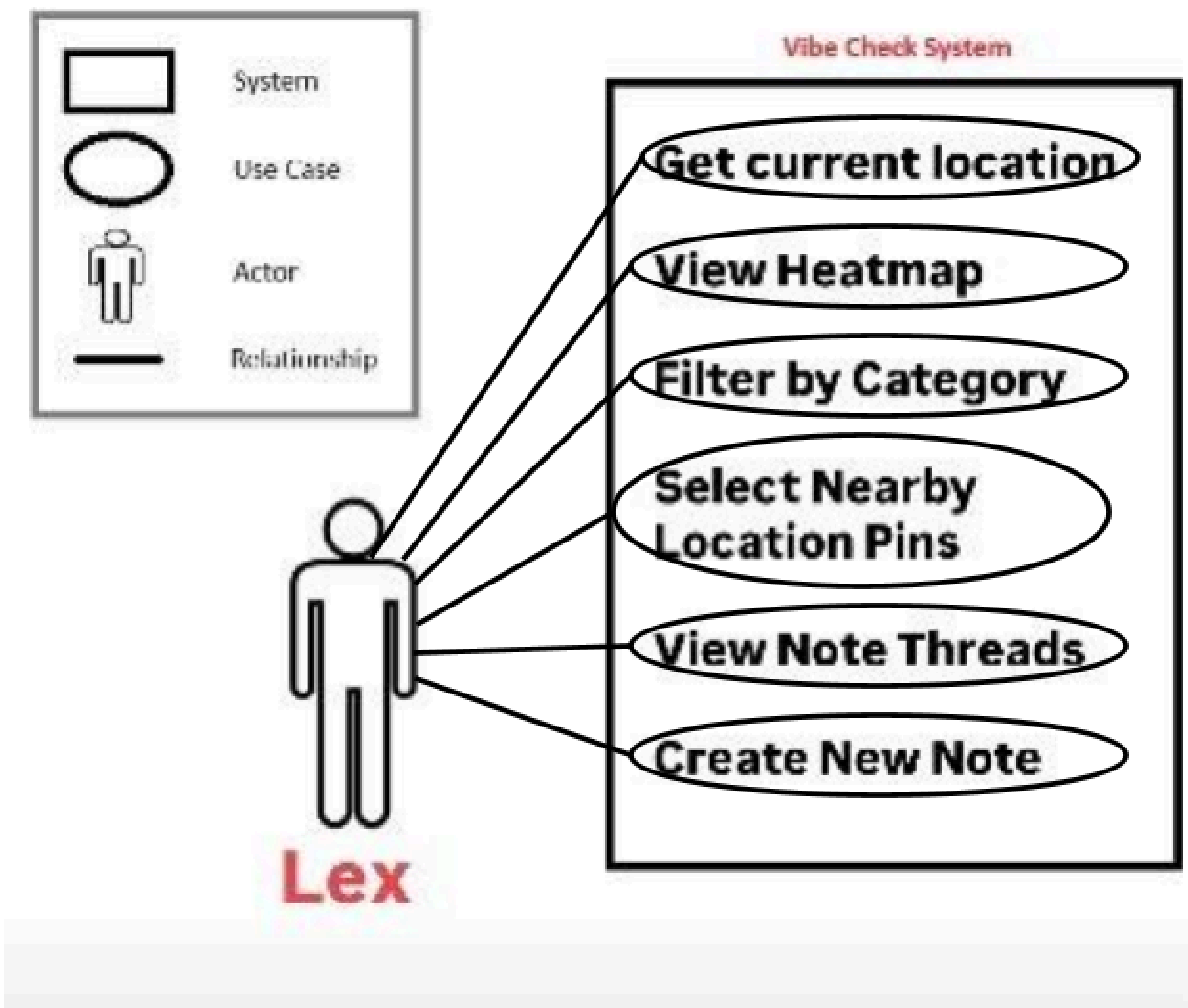
Assumptions:
- Lex has access to the internet with a mobile device
- Lex has location enabled services turned on
- There is existing reviews and comments on location

Use Case: 3

Lex is a hybrid worker, but prefers to work in a more work oriented environment during her times she is not in the office. She looks for places to work, but is unable to find something that she is sure about which matches the environment she is looking for. She finds an app called Vibe Check. From the app, she is able to get a better insight on where she goes and the aesthetic it goes for under a tab of quiet places. Lex is happy to see the specificity of vibes of a place through the check-in boards, as well as the history of recent activities at a certain place. After reviewing multiple locations' look and check-in boards comments, she sees a perfect spot with a photo attached and a comment saying “I found this hidden spot nearby my college campus, it’s really quiet but very aesthetically pleasing to work in”. After visiting the location she is pleased with her experience. She reacts to the other person's comment and shares it with the community for people who are looking for a similar experience.

Benefits for Lex:

- Lex is able to find quiet places around her to focus on her work
- Lex is able to find an environment that inhibits her work style
- Lex saves time in finding places instead of visiting places that might have unavailable seating



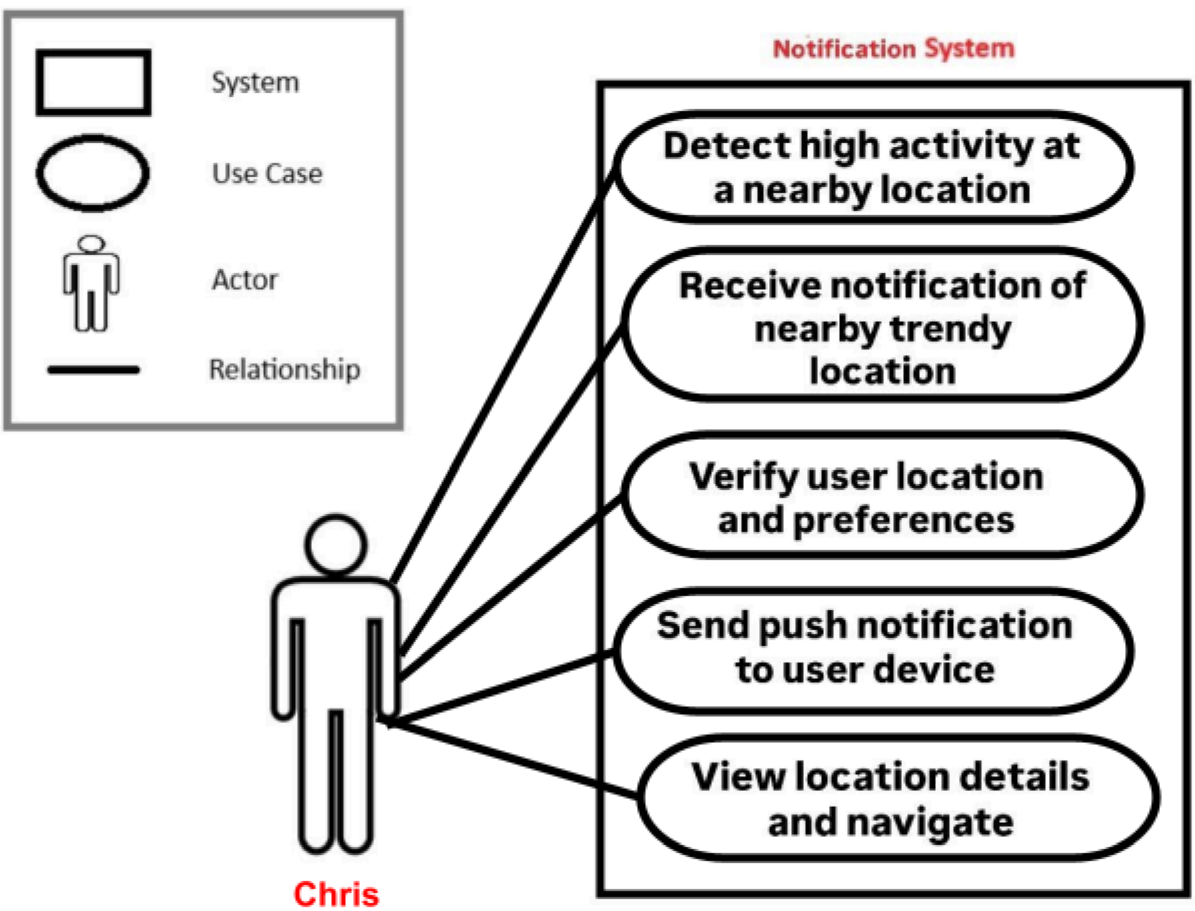
Notified of Nearby Trendy Location

Actors:
Chris (Registered User), Notification

Assumptions:
- Chris has access to the internet with a mobile device
- Chris has location enabled services turned on

Use Case: 4
Chris is at his house trying to make Friday night plans, he likes to make new friends so when he goes out he prefers to go somewhere with a lot of people to meet. He gets a notification from the Vibe Check app. The notification tells him that a venue near him is bustling tonight. Chris is happy to have quickly found a high energy spot to spend his evening.

Benefits for Chris:
- Chris is able to find a lively place to meet new people
- Chris saves time looking for places or showing up to places that are dead



Reviewing Personal Store

Actors: Carlos (Registered User, Business Owner), Vibe Check (System)

Assumptions:

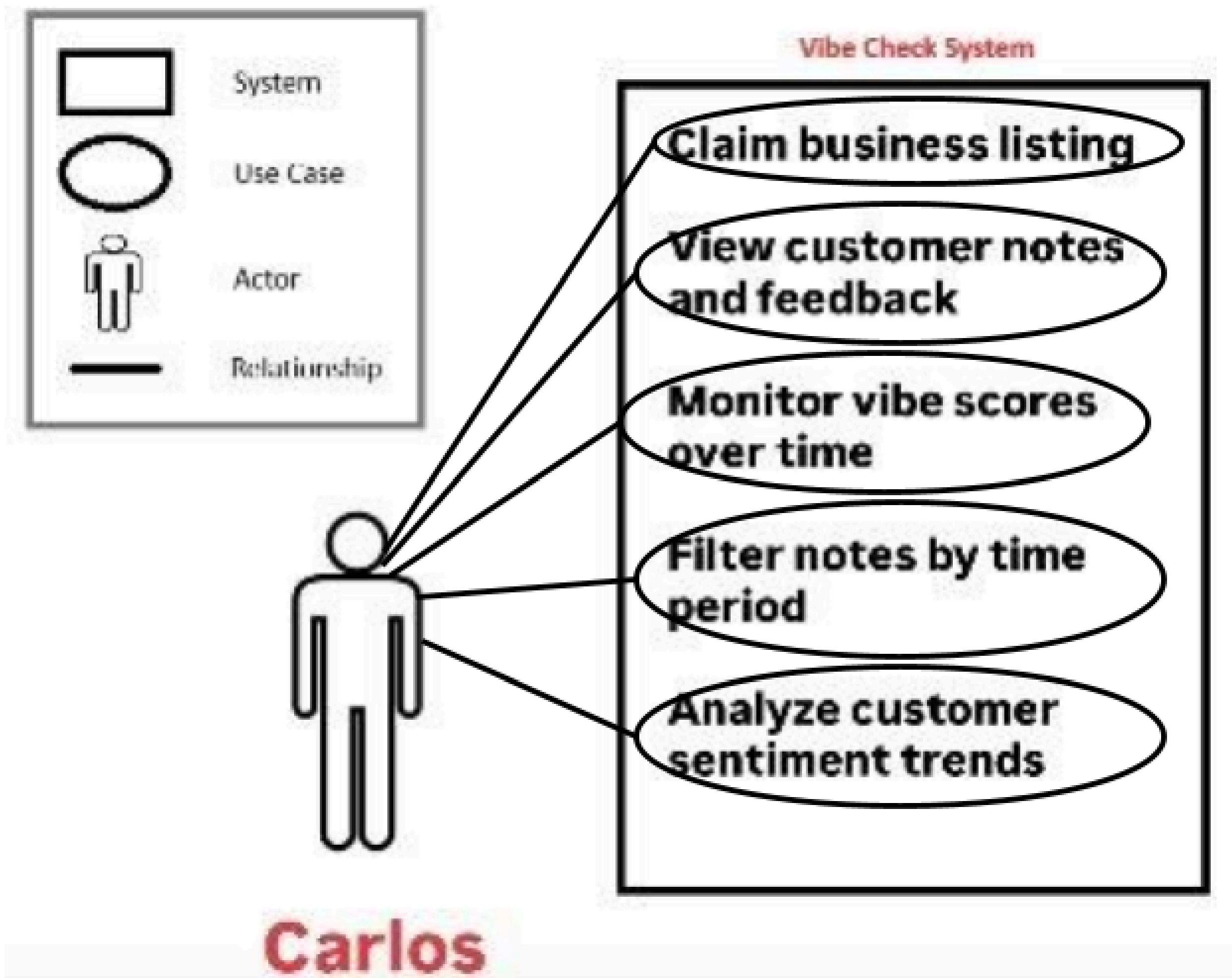
- Carlos claimed his business listing on Vibe Check
- Carlos has access to the app
- There is existing reviews and comments on location

Use Case: 5

Carlos owns a mid-sized coffee shop and wants to understand how customers are perceiving the atmosphere during different times of the dat. He monitors his location on Vibe Check throughout the day. During the rushes he is able to read the notes/comments of his cafe and reads notes that say "great music at low volumes, and a pretty comforting place to study at". He also is able to read other notes at certain times saying "It was too busy and loud during our visit". He lowers his volume and increases seating at more busier hours due to the feedback and optimizes the way he works through crowdsourced comments

Benefits for Carlos:

- Carlos is able to receive feedback and constructive criticism on his business.
- Carlos is able to make adjustments to customer sentiment.
- Carlos builds a stronger reputation catering to customers over time.



Community Gatherings and Meetings

Actors: Abel (Registered User, Community Contributor), Vibe Check (System)

Assumptions:

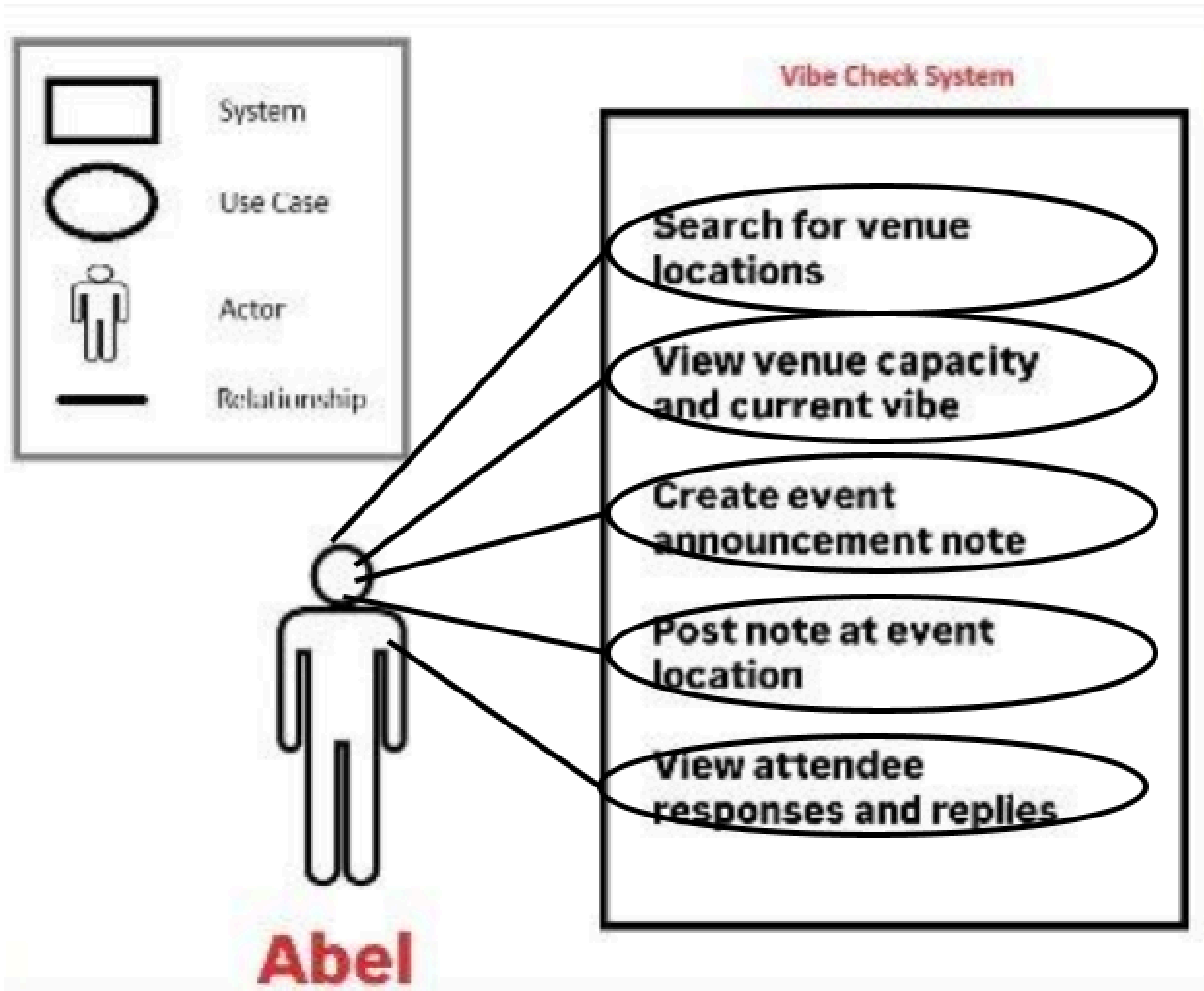
- Abel has access to the internet with a mobile device
- Abel has location enabled services turned on
- There is an accessible bulletin board/comment section to advertise

Use Case: 6

- Abel wants to support his community event at a local cafe to bring in new members, but has trouble advertising it and getting members to join. He remembers that Vibe Check had a comment section and posts real-time. He looks for a spacious place to host his meet up and he posts “I am hosting a small mixer here if anybody wants to talk, make new friends and is passing by”. He creates his post, and is happy to see people introducing themselves. Abel is happy that he was able to send real-time information and meet with new people who have similar interests. He recommends people to download the app to keep up with more community events around the area and his as well.

Benefits for Abel:

- Abel is able to advertise his event in real time to crowd source users with similar interests.
- Abel is able to gather new members for his community.



Meeting up with Friends

Actors: Sue (Registered User) Friends

Assumptions:

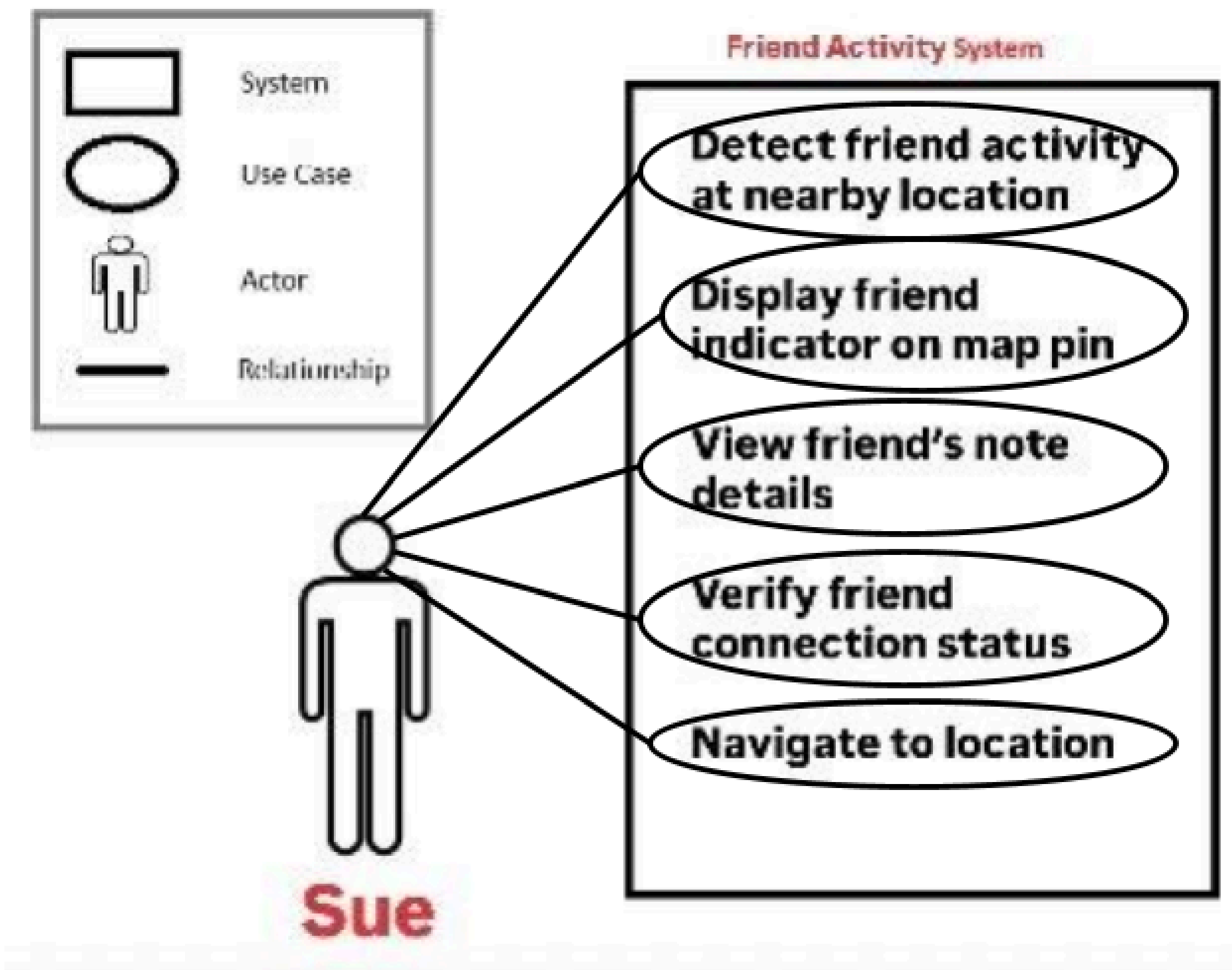
- Sue has access to the internet with a mobile device
- Sue has location services enabled on her device
- Sue has at least one accepted friend connection on the platform
- Sue's friend has recently left a note at a nearby location

Use Case: 7

- Sue is deciding where to spend her Saturday evening. She's in the mood for a relaxed atmosphere but doesn't want to go somewhere alone. She opens Vibe Check and views the heatmap of her area. As she browses, she notices a special icon on a café pin indicating that her friend Maya recently left a note there. Sue taps on the pin and reads Maya's note: "Cozy corner spot available, live acoustic music tonight. Not too crowded!" Sue also sees that Maya posted this just 15 minutes ago, meaning she's likely still there. Sue feels excited knowing she can join a friend rather than showing up somewhere solo. She sends Maya a quick text message saying she's on her way, then checks out a few other notes at the café to confirm it matches the vibe she's looking for. Other users confirm the relaxed atmosphere and available seating. Sue heads to the café, meets up with Maya, and enjoys a great evening with live music in good company. Before leaving, Sue adds her own note to the location saying "Great night here with a friend! Highly recommend the corner seats " and tag the vibe level as "Moderate."

Benefits for Sue:

- Sue discovers where her friend is in real-time without needing to text and coordinate
- Sue makes a confident decision about where to spend her evening knowing she won't be alone
- Sue benefits from both her friend's firsthand recommendation and the broader community's real-time updates
- Sue strengthens her social connections by spontaneously meeting up with friends



Fun With Badges

Actor: Bruno (Registered User) Badges

Assumptions:

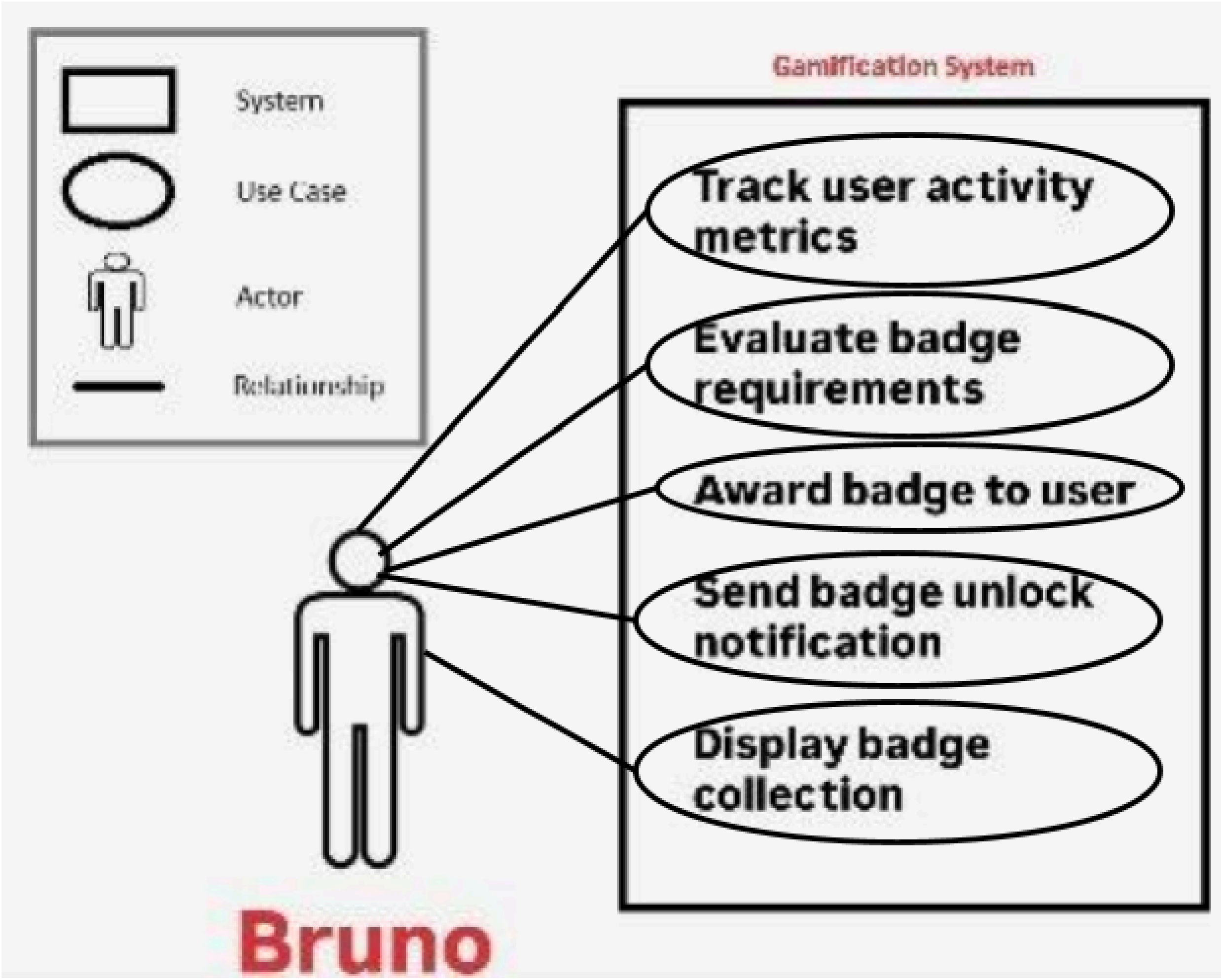
- Bruno has access to the internet with a mobile device
- Bruno has location services enabled on his device
- Bruno has been actively using Vibe Check and posting notes

Use Case: 8

- Bruno is a college student who has gotten into the habit of stopping by his favorite coffee shop every morning before class. Each time he visits, he leaves a quick note on Vibe Check sharing whether it's busy, quiet, or if there are any good seats available. Other students have found his updates helpful and often react positively to his notes. One Thursday morning, Bruno posts his note: "Plenty of seats by the windows today, barely any line " and selects the vibe level as "Quiet." As soon as he submits the note, a notification pops up with a badge icon and celebratory animation: "🏆Badge Unlocked: Coffee Connoisseur - Posted notes at coffee shops 10 times!" Bruno taps on the notification and views the badge details. He sees that he's earned other badges too: "First Drop" for his first-ever note and "Early Bird" for posting 5 notes before 9 AM. He also notices he's halfway to earning the "Local Legend" badge, which requires 50 total notes. Feeling motivated, Bruno decides to keep contributing. He starts posting notes when he visits other locations throughout the day, the library, the gym, and a taco spot near campus. By the end of the week, he's earned the "Campus Explorer" badge for posting notes at 5 different campus locations.Bruno checks his profile and sees all his earned badges displayed. He feels a sense of accomplishment and community contribution. He also notices that other users can see his badges, which gives his notes more credibility. When new users see Bruno has the "Coffee Connoisseur" badge, they trust his café recommendations more.

Benefits for Bruno:

- Bruno feels recognized and rewarded for his consistent contributions to the community
- Bruno is motivated to continue posting helpful notes at various locations
- Bruno's badges build his credibility, making his notes more trustworthy to other users
- Bruno discovers new locations as he explores different areas to earn location-specific badges
- Bruno experiences gamification that makes using the app more engaging and fun



Main Data Items and Entities

User

A user is a person who uses the application to view nearby locations and interact with content. A User can create Notes and can reply to or react to existing Notes.

Location

A Location represents a physical place shown on the map (e.g., café, library, venue, park). Notes are tied to a specific Location to ensure that updates are relevant to that place.

Note

A Note is a short, location-locked post created by a User to describe what is happening at a Location in real time. Each Note has a timestamp and an expiration time so that outdated information is automatically removed.

Category Tag

A Category Tag is a label used to classify a Note or Location (e.g., restaurant, study spot, venue line, parking). Category Tags allow users to filter and view relevant content more efficiently.

Reply

A Reply is a response to a specific Note that provides additional context or updated information. Each Reply belongs to one Note.

Reaction

A Reaction represents a quick feedback action (such as an upvote or similar interaction) that a User can apply to a Note. Reactions help indicate which notes are useful or accurate.

Heatmap Activity

Heatmap Activity is a visual representation of recent activity in a geographic area. It is generated from recent Notes and user interactions within a specific time window and helps users quickly assess how active an area is.

User Profile

A user profile stores user preferences such as notification settings, default radius, preferred categories. It tracks user contribution stats (for future gamification).

Notification

A notification represents a system notification (trending location nearby, etc.)

Friend

A Friend Connection represents a mutual relationship between two Users. It allows Users to see each other's Notes on the map and receive notifications when a friend leaves a Note nearby.

Vibe Score

A Vibe Score is a quick, standardized rating a User can apply to a Location through their Note, similar to Google Maps' thumbs system but framed around atmosphere. Instead of stars, users rate using descriptors (e.g., Quiet, Moderate, Buzzing) or a simple scale.

Badge

A Badge is an achievement awarded to a User for meeting specific contribution milestones (e.g., leaving 10 notes, visiting 5 cafes, being a top contributor in an area).

Logical Relationships

- A User can create many Notes.
- A Note belongs to one Location.
- A Note can have many Replies.
- A User can create many Replies.
- A User can create many Reactions, and each Reaction applies to one Note.
- A Location can have many Notes over time.
- Heatmap Activity is derived from recent Notes and interactions.
- Each User has one Profile
- Profile stores preferences and activity history
- A Notification belongs to one User
- A Notification references one Note or Location
- A User can have many Friend Connections
- A Friend Connection belongs to exactly two Users
- A Friend Connection enables visibility of each User's Notes to the other
- A Vibe Score belongs to one Note
- A Note has at most one Vibe Score
- A Location's aggregate Vibe Score is derived from recent Note ratings
- A User can earn many Badges
- A Badge can be earned by many Users
- Badges are awarded automatically when a User meets the threshold

Functional Requirements

1. Authentication

- FR 1.1: The user shall be able to create an account with an email and password.
- FR 1.2: The user shall be able to log in and log out.
- FR 1.3: The user shall be required to be authenticated in order to create notes, replies, or reactions.

2. Location

- FR 2.1: The user shall be able to grant permission for the app to access their current location.
- FR 2.2: The user shall be able to view nearby locations on the map.
- FR 2.3: The user shall be able to select a specific location from the map.

3. Heatmap

- FR 3.1: The user shall be able to view a heatmap showing activity in the nearby area.
- FR 3.2: The user shall be able to filter the heatmap by category.
- FR 3.3: The user shall be able to zoom in and out of the map.

4. Notes

- FR 4.1: The user shall be able to view notes related to a specific location.
- FR 4.2: The user shall be able to view replies to a note.
- FR 4.3: The user shall be able to reply to notes posted by other users.
- FR 4.4: The user shall be able to delete their own reply.
- FR 4.5: The user shall be able to view reactions to a note.
- FR 4.6: The user shall be able to react to notes posted by other users.
- FR 4.7: The user shall not be able to react multiple times to the same note.
- FR 4.8: The user shall be able to remove a reaction.
- FR 4.9: The user shall be able to post a note at a specific location.
- FR 4.10: The user shall be able to compose text content when creating a note.
- FR 4.11: The user shall be able to edit their own note.
- FR 4.12: The user shall be able to delete their own note.
- FR 4.13: The user shall not be able to post from outside the defined radius
- FR 4.14: The user shall be shown an error message if posting fails.
- FR 4.15: The user shall be able to view notes in a card format resembling a letterboxed review, showing vibe level, note content, timestamp, reaction count, and reply count.
- FR 4.16: The user shall be able to visually distinguish between notes posted by friends and notes posted by other users on the map.
- FR 4.17: The user shall be able to view the username or anonymous handle associated with each note.
- FR 4.18: The user shall be able to view the time elapsed since a note was posted.
- FR 4.19: The user shall be able to view the remaining time before a note expires.

5. Notification

- FR 5.1: The user shall be able to receive a notification when a trendy location is nearby.
- FR 5.2: The user shall be able to enable or disable notifications in settings.

6. Search/Filter

- FR 6.1: The user shall be able to search for locations by name or category.
- FR 6.2: The user shall be able to filter search results to a radius.
- FR 6.3: The user shall be able to filter search results by vibe level.

7. Vibe Score

- FR 7.1: The user shall be able to select a vibe level (Dead, Quiet, Moderate, Busy, Buzzing) when creating a note.
- FR 7.2: The user shall be able to view the aggregate vibe score of a location based only on non-expired notes.
- FR 7.3: The user shall see a fallback message if insufficient data exists for the vibe score.
- FR 7.4: The user shall be able to view vibe scores visually using descriptive labels rather than numerical star ratings.
- FR 7.5: The user shall be able to view an updated vibe score for a location when new notes are added or expired notes are removed.
- FR 7.6: The user shall be able to filter locations on the map by vibe level.

8. Friends

- FR 8.1: The user shall be able to send a friend request to another registered user.
- FR 8.2: The user shall be able to accept or decline an incoming friend request.
- FR 8.3: The user shall be able to remove a friend from their friends list.
- FR 8.4: The user shall be able to view a list of their current friends.
- FR 8.5: The user shall be able to see notes left by their friends on the map, visually distinguished from notes by non-friends.
- FR 8.6: The user shall be notified when a friend leaves a note at a nearby location.
- FR 8.7: The user shall be able to enable or disable friend activity notifications in settings.
- FR 8.8: The user shall be able to block another user, preventing them from viewing their notes or sending friend requests.
- FR 8.9: The user shall not be able to view notes posted by users they have blocked.
- FR 8.10: The user shall be able to unblock a previously blocked user.
- FR 8.11: The user shall be able to see an indicator icon on map pins where a friend has recently left a note.

9. Badge

- FR 9.1: The user shall be automatically awarded badges when they meet defined contribution thresholds.
- FR 9.2: The user shall be able to view all badges they have earned on their profile.
- FR 9.3: The user shall be able to view all available badges and the requirements to earn them.
- FR 9.4: The user shall be notified when they earn a new badge.
- FR 9.5: The user shall be able to track the number of notes they have posted toward badge thresholds.
- FR 9.6: The user shall be able to track the number of unique location types (e.g., cafes, venues, parks) they have posted notes at.
- FR 9.7: The user shall be able to view the number of reactions their notes have received toward badge thresholds.
- FR 9.8: The user shall be able to display earned badges on their public profile.

10. Profile Management

- FR 10.1: The user shall be able to edit their username.
- FR 10.2: The user shall be notified if the username they want to change to is unavailable.
- FR 10.3: The user shall be able to change their password.
- FR 10.4: The user shall be able to upload a profile picture.
- FR 10.5: The user shall be able to delete their account.
- FR 10.6: The user shall be able to view their posting history.

11. Moderation

- FR 11.1: The user shall be able to report a note for inappropriate content.
- FR 11.2: The user shall be able to provide a reason for reporting a note.
- FR 11.3: The user shall be able to report a reply for inappropriate content.
- FR 11.4: The user shall be able to provide a reason for reporting a reply.

Non Functional Requirements

1. Performance

- NF 1.1: The system shall load the heatmap of the user’s vicinity within 3 seconds.
- NF 1.2: The system shall update a location's aggregate vibe score within 5 seconds of a new note being submitted.
- NF 1.3: The system shall load a user's friends' map activity within 3 seconds of opening the map view.
- NF 1.4: The system shall deliver push notifications within 30 seconds of a triggering event (e.g., friend nearby, trending location).

2. Reliability

- NF 2.1: The system shall keep notes, replies, and reactions persistent until their preset expiration time.
- NF 2.2: The system shall not send duplicate notifications for the same event to the same user.

3. Security

- NF 3.1: The system shall securely store user passwords using hashing and salting.
- NF 3.2: The system shall not expose the user’s exact location to other users.

4. Usability

- NF 4.1: The system shall warn the user of missing or denied location access and offer a fallback solution.
- NF 4.2: The system shall use high-contrast colors and readable font sizes conforming to WCAG 2.1 AA standards.

5. Scalability

- NF 5.1: The system shall support an increasing number of notes as the user base grows.
- NF 5.2: The system shall maintain consistent performance under peak activity periods.

6. Compliance

- NF 6.1: The system shall request user permission before accessing location services.
- NF 6.2: The system shall provide users with the ability to request deletion of their personal data.

7. Maintainability

- NF 7.1: The system shall follow a consistent coding standard and project structure across the frontend and backend.
- NF 7.2: The system shall maintain separation between frontend, backend and database components to support modular development.
- 7.3: The system shall include API documentation for all major backend endpoints.

8. Privacy

- NF 8.1: The system shall allow users to set their profile visibility to public or friends-only.

- NF 8.2: The system shall not share a user's precise location history with other users, including friends.
- NF 8.3: The system shall allow users to post notes anonymously, hiding their username from public view.

9. Data Retention

- NF 9.1: The system shall retain user badge history indefinitely unless the user deletes their account.
- NF 9.2: The system shall automatically delete expired notes after their expiration time.

Competitive Analysis

Competitive Features Comparison

Key comparison area	**Google Maps**	**Yelp**	**Snapchat Snap Map**	**Swarm (Foursquare)**	**Waitz (Occuspace)**	**Vibe check**
Core value: real-time "vibe" decisioning (quiet/loud/crowded/active)	Medium (busyness, not "why")	Medium (reviews lag)	Low (unstructured)	Low (check-ins)	Medium (busyness only)	Strong, purpose-built
Freshness control (keeps info current)	Medium (live busyness sometimes)	Low (reviews persist)	Low (not freshness-driven)	Low	Medium (sensor-based, no expiry)	Strong (expiration)
Trust & relevance (credible for this exact place)	Medium (algorithmic)	Medium (varies by reviewer)	Low (context unclear)	Medium (presence implied)	Medium (sensor-based)	Strong (location-locked + confirmations)
Discovery UX (scan area fast)	Strong (map + popular times)	Medium (search/reviews)	Strong (map browsing)	Medium	Low (campus only)	Strong (heatmap + filters)
Content density & coverage (works everywhere)	Very High	High	Very High	Low-Med	Low	Low early (cold start)
Distribution & habit (default app users open)	Very High	High	Very High	Low-Med	Low	Low (new)
Operational risk	Medium	Medium	High	Medium	Low	High

(spam/trolling/moderation)						

Competitive Features Table

Feature	**Vibe Check**	**Google Maps**	**Yelp**	**Snap Map**	**Swarm**	**Waitz**
Real-time, expiring vibe notes	++	-	-	-	-	-
Location-locked posting (must be near)	++	-	-	-	+	+
Heatmap based on recent activity	++	-	-	+	-	++
Structured vibe categories (quiet/loud/crowded)	++	+	+	-	-	-
Works well with low user base	+	++	++	++	+	++

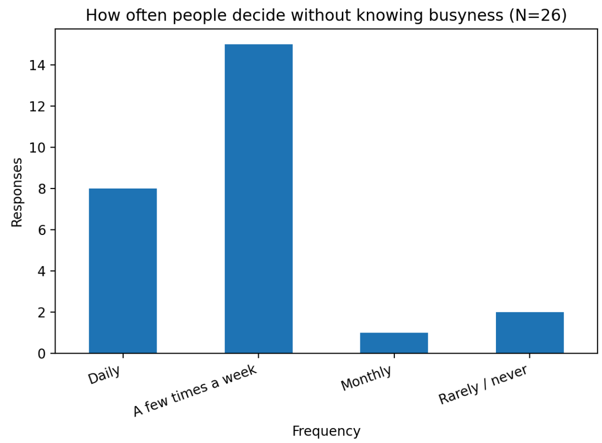
Target Audience Validation Summary

User Survey: https://docs.google.com/forms/d/1PQx8zkkx3Yrz9ZkarF85jIr0L9O-EvwaWALYa40v2oE

We surveyed *potential users* to validate demand for Vibe Check and test whether our “unique” features (freshness and trust) match real needs.

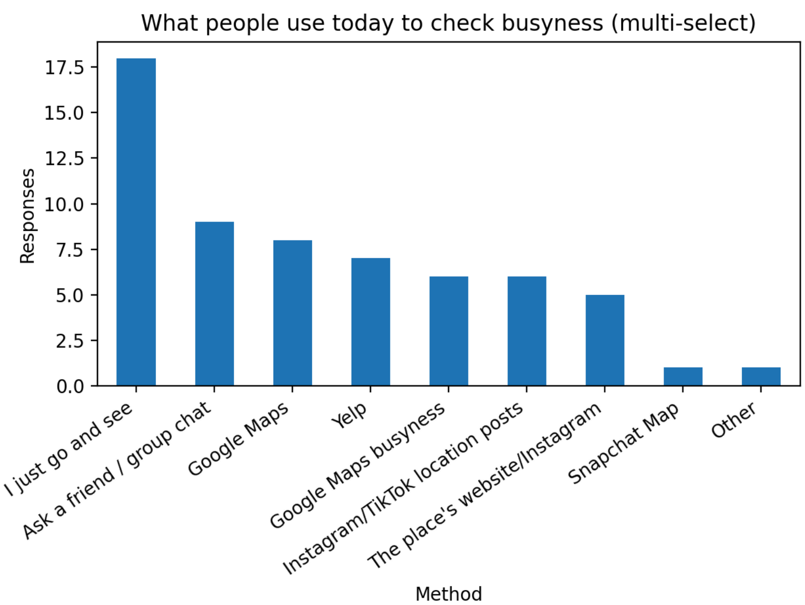
Who responded (target fit):
- Students: (46.2%) *(Student + University student + Graduate student)*
- Full-time in-person workers: (26.9%)
- Other groups: (26.9%) *(hybrid/remote, tourist, other)*

Problem frequency (need is real): People decide where to go without knowing busyness daily or a few times/week: (88.5%) *(Daily: 30.8%; Few times/week:57.7%)*



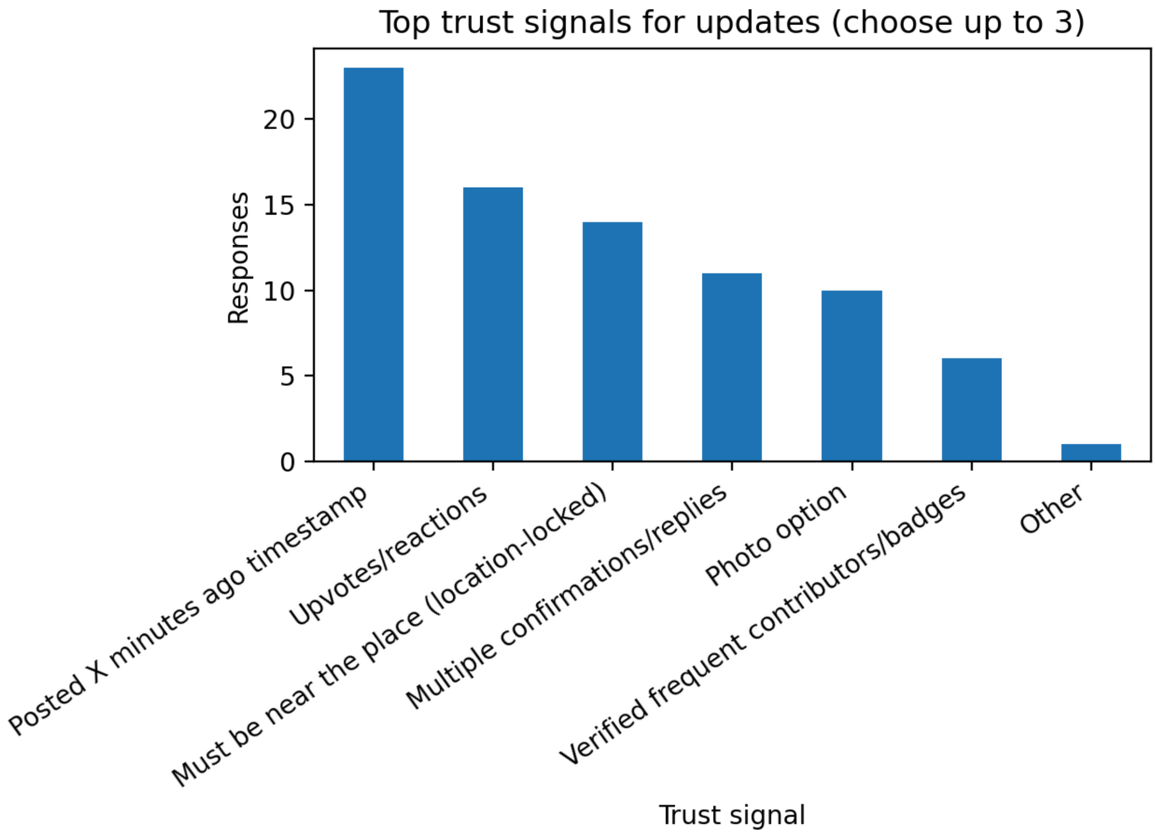
Why current solutions fail:

- Top pain points: "busy but not why" (42.3%), "outdated by arrival" (42.3%), not helpful for venues (38.5%), not helpful for study spots (34.6%).
- Most common current behavior: "I just go and see" (69.2%)



Feature validation (our differentiators are supported):

- Expiring updates increases trust: (76.9%) (Yes definitely/somewhat).
- Location-locked posting increases trust: (96.2%).
- Top trust signals: timestamp (88.5%), reactions (61.5%), location-lock (53.8%)



Competitive Analysis Summary

The competitive comparison reveals that existing tools either provide **broad coverage without context** or **context without real-time freshness**. Google Maps is effective for quick discovery and often provides a general busyness signal, but it does not reliably explain *why* a location feels unsuitable (e.g., “line around the block,” “group projects are loud,” “no seating available”). Yelp provides useful historical opinions and ratings, but its content is not designed for time-sensitive decisions because reviews persist long after conditions change. Social map products such as Snapchat Snap Map (and check-in style apps like Swarm) can indicate that activity is happening, but the information is typically unstructured and not optimized to answer “Should I go here right now for my specific need?” Waitz solves part of the problem by providing real-time crowd levels, but it still lacks user-provided situational context and does not support short-lived, location-locked notes that capture atmosphere details.

Based on these findings, Vibe Check’s differentiation is not simply “real-time,” but **real-time and decision clarity**. Vibe Check combines:

1. **Location-locked posting** to improve credibility (users must be near the location),
2. **Expiring notes** to enforce freshness and reduce outdated content, and
3. A **heatmap + category filters** to make discovery fast and goal-oriented (e.g., quiet study spots vs lively venues).
   Together, these features directly address the core user pain point identified in our use cases: **avoiding wasted trips and choosing places that match immediate needs (quiet, quick service, lively atmosphere, etc.)**.

Project Readiness Checklist

- [x] ~~Meeting schedule: Every Monday at 7pm, Every Wednesday at 7pm (optional)~~

  https://www.notion.so/9d029572065c4fcf9595b39325b1a3d6?v=ca5e021bf02b47b98fb274dba16051ec&source=copy_link

- [x] ~~Roles assigned:~~
  -Harry Kakadiya: Team Lead
  - Kaitlyn Ashby: Github master, Technical writer
  - Rahul Srinath: Software Architect
  - Amulya Sagi: Backend Lead
  - Aljhay Soriano: Scrum Master
- [x] ~~Tools selected (all teammates have basic experience, those of us with more experience can help those with less and point them to documentation):~~
  Cloud: AWS (Free Tier), t2.small — 1 vCPU / 2 GB RAM
  OS: Ubuntu Server 22.04 LTS
  Web Server: Nginx 1.24
  Backend: Node.js 20.x, Express.js 4.x (JavaScript/TypeScript)
  Frontend: React 18 (Vite), Tailwind CSS
  Database: PostgreSQL 15.x with PostGIS
  AI (Optional): OpenAI API
  Tools: Leaflet.js, OpenStreetMap, Socket.io
- [x] ~~Repository organized correctly (Kaitlyn in charge of this as github master)~~

High Level Architecture and Technologies

All approved by instructor

- Cloud: AWS (Free Tier), t3.micro
- OS: Ubuntu Server 24.04 LTS
- Web Server: Nginx 1.24
- Backend: Node.js 20.x, Express.js 4.x
- Frontend: React 18 (Vite), Tailwind CSS
- Database: PostgreSQL 15.x with PostGIS
- Tools: Leaflet.js, OpenStreetMap, Socket.io
- AI (optional): OpenAI API

Team Contributions

1. Harry Kakadiya — Team Lead
Version 1 Contributions:
Revision table: v2
Table of contents
Main data entities: User Location Note Category Tag Reply Reaction
Heatmap Activity

Version 2 Contributions:
Designed and configured the project database
Set up database credentials and schema
Logical Relationships: 1-7
Non-functional Requirements: 7.1
Competitive Analysis: Waitz
Project Readiness checklist
High level architecture and technologies

2. Kaitlyn Ashby — GitHub Master, Technical Writer
Version 1 Contributions:
Revision table: v1
Main use cases: Notified of Nearby Trendy Location
Main data entities: User Profile
Notification
Version 2 Contributions:
Reviewed Github Repo
Structure Reviewed final
M1.pdf contributions.md
Use cases: Meeting up with Friends Fun With Badges
Main data entities: Friend Vibe Score Badge
Logical Relationships: 8-20
Functional requirements: 4.15-6.1 7.1-7.2 7.4-8.8 8.11-9.8
Nonfunctional requirements: 1.2-1.4 2.2 4.2 8.1-9.1

3. Rahul Srinath — Software Architect
Version 1 Contributions:
Executive summary
Main use cases: Checking a Busy Restaurant Before Committing
Functional Requirements: 1.1-4.14 6.2-6.3 7.3 8.9-8.10 10.1-11.4
Non-functional requirements: 1.1 2.1 3.1-4.1 5.1-6.2 7.2-7.3 9.2
Version 2 Contributions:
Implemented the About page

4. Amulya Sagi — Backend Lead Version 1 Contributions:
Main use cases: Finding a Quiet Study Spot on Campus

Version 2 Contributions:
Competitive Analysis
Configured and deployed the web server

5. Aljhay Soriano — Scrum Master
Version 1 Contributions:
Main use cases: Reviewing Places by Popularity
Version 2 Contributions:
Main use cases: Reviewing Personal Store Community Gatherings and Meetings
Set up and configured AWS EC2 instances
Managed SSH access and key pair configuration
Configured security groups and networking (ports 22, 80, 443)

Member	Score
Harry	5/5
Kaitlyn	5/5
Rahul	5/5
Amulya	5/5
Aljhay	5/5

Completed before class
https://docs.google.com/forms/d/e/1FAIpQLSdJvLYO2UiAZyPgD6BVVzhdtwVGLYfRAwXmfTN9UTm12THizw/viewform